course code:- CSA5107

course :- Cryptography CIA-1

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1. Caesar cipher and Vigenere cipher

Given,

Plaintext : MEETATNOON

Caesar shift key : 7

Vigenere key : SUN

Numeric conversion: A=0, B=1 - - - Z=25

Caesar cipher Encryption

Formula :- $C = (P+7) \bmod 26$

convert each letter into its numeric value and add 7.

Plaintext	P value	calculation	C value	ciphertext
M	12	(12+7) mod 26	19	T
E	4	(4+7) mod 26	11	L
E	4	(4+7) mod 26	11	L
T	19	(19+7) mod 26	0	A
A	0	(0+7) mod 26	7	H
T	19	(19+7) mod 26	0	A
A	19	(19+7) mod 26	0	A
T	19	(19+7) mod 26	20	U
N	13	(13+7) mod 26	21	V
O	14	(14+7) mod 26	21	V
O	14	(14+7) mod 26	20	U
N	13	(13+7) mod 26	20	U

ciphertext :- TLLAHAAVVU

Caesar cipher Decryption :-

Formula :- $P = (C-7) \bmod 26$

ciphertext	C value	calculation	Pvalue	Plainte
T	19	(19-7) mod 26	12	M
L	11	(11-7) mod 26	4	E
L	11	(11-7) mod 26	4	E
L	0	(0-7) mod 26	19	T
A	7	(7-7) mod 26	0	A
H	0	(0-7) mod 26	19	T
A	20	(20-7) mod 26	13	N
U	21	(21-7) mod 26	14	O
V	21	(21-7) mod 26	14	O
U	20	(20-7) mod 26	13	N

Decrypted text : MEETATNOONN

Vigenere cipher Encryption

Given

plaintext : MEETATNOONN

key : SUN

Repeated key : SUNISUNISUNIS

Formula :-

$$C = (P+K) \bmod 26$$

Plaintext	P value	Key	Key value	calculation	cipher
M	12	S	18	(12+18) mod 26	E
E	4	U	20	(4+20) mod 26	Y
E	4	N	13	(4+13) mod 26	R
E	19	S	18	(19+18) mod 26	L
T	0	U	20	(0+20) mod 26	U
A	19	N	13	(19+13) mod 26	G
T	13	S	18	(13+18) mod 26	F
N	14	U	20	(14+20) mod 26	I
O	14	N	13	(14+13) mod 26	B
O	13	S	18	(13+18) mod 26	F
N					

ciphertext: EYRLUGFIBF

Vigenere cipher Decryption

Formula:- $P = (C - K) \bmod 26$

ciphertext	key	Decryption result
E	S	M
Y	U	E
R	N	E
L	S	T
U	U	A
G	N	T
F	S	N
I	U	O
B	N	O
F	S	

Decrypted plaintext: MEETATNOON

2> Hill cipher Encryption and Decryption

Given,

=> Plaintext : MATH

=> Key Matrix : $K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

Comparison :-

Feature	caesar	Vigenere
Key space	25 shift	depends on key length
Repeated letter	same output	different output possible
Security	Very weak	stronger than caesar
Attack	Brute force	Frequency and key-length analysis

Recommendation :- Vigenere is better for educational demonstrate because it uses multiple letter shifts.

conclusion :- Neither cipher is suitable for modern confidential data. AES-256-GCM Provides strong encryption, authentication and integrity protection.

Encryption :- convert letters into numbers (A=0 to Z=25)

M=12, A=0, T=19, H=7

divide into pairs:

$$MA = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \quad TH = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

Formula :— $C = KP \bmod 26$

For MA: $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} 36 \\ 60 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 8 \end{bmatrix}$

10=K and 8=I

For TH:

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} 71 \\ 130 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 0 \end{bmatrix}$$

19=T and 0=A

ciphertext: KITA

Decryption :—

calculate the determinant:

$$|K| = (3 \times 7) - (2 \times 5) = 11$$

since gcd(11,26) = 1, the inverse exists

The modular inverse of 11 is 19.

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

using the inverse matrix:

$$P = K^{-1} C \bmod 26$$

The decrypted plaintext is MATH

Invalid key Matrix :-

Consider:

$$K = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

Determinant:

$$(2 \times 5) - (3 \times 4) = -2$$

Since $\gcd(-2, 26) = 2$, the inverse does not exist

∴ This matrix is not a valid Hill cipher key.

Conclusion :- The Hill cipher encrypts and decrypts MATH using a valid key matrix. However, it is vulnerable to known-Plaintext attacks.

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3) a) AES-256-GCM and RSA Comparison

AES-256-GCM	RSA
* symmetric algorithm	Asymmetric algorithm
* Uses one shared secret key	Uses Public and private keys
* Very fast	comparatively slow

* Suitable For 1GB MRI Files
* provides authentication

→ suitable for small encryption keys
→ RSA encryption alone does not provide authentication.

Scalability :-

For 200 clinics communicating pairwise with shared symmetric keys :

$$\frac{200 \times 199}{2} = 19,900 \text{ keys}$$

* In a central-server model, 200 clinic-to-server shared keys are needed.
* with RSA, each clinic and server maintain its own Public-private key pair.
* Digital certificates are used to verify the ownership of public keys.

Conclusion :- AES is suitable for large MRI files, while RSA is suitable for secure key exchange.

b) Hybrid Encryption workflow

Sending clinic :

* validate the central server's digital certificate.
* Generate a random 256 bit Data Encryption key (DEK) and unique nonce.

* Encrypt the MRI file using AES-256-GCM with ass data.
* Encrypt the DEK using the server's RSA Public key with OAEP Padding.

Receiving Server :-

* Use the RSA private key to unwrap the DEK.
* Use AES-GCM to decrypt the MRI files.
* Verify the authentication tag.
* Reject file if authentication fails.

Secure Key Management :-

* store keys in HSM/KMS
* Use nonexportable keys and role-based access
* Apply MFA and audit logging
* Maintain secure backup and recovery.

Key Rotation :-

* Generate a separate DEK for each file.
* Periodically rotate the key-encryption key.
* Re-wrap stored DEK with the new key.
* Retire old keys securely.

Conclusion :- Hybrid encryption provides secure efficient data transfer.

RSA Encryption and Decryption

Given,

$p = 11$

$q = 19$

$e = 7$

$M = 5$

calculate n :-

$$n = p \times q = 11 \times 19 = 209$$

calculate Euler's Totient

$$\phi(n) = (p-1)(q-1)$$

$$= (11-1)(19-1)$$

$$= 10 \times 18$$

$$= 180$$

calculate private key

$$ed \bmod 180 = 1$$

$$7 \times 103 = 721$$

$$721 \bmod 180 = 1$$

$\therefore d = 103$

Public key : $(7, 209)$

Private key : $(103, 209)$

Encryption :-

Formula :- $C = M^e \bmod n$

$$C = 5^7 \bmod 209$$

$$C = 168$$

∴ Encrypted Message = 168

Decryption :-

Formula : $M = C^d \bmod n$

$$M = 168^{103} \bmod 209$$

$$M = 5$$

∴ Decrypted Message = 5

Security Limitations

* Small RSA keys can be broken easily
* Textbook RSA is deterministic
* Modern RSA uses large key sizes such as 2048 bits or more.
* RSA - OAEP is recommended for encryption

Conclusion :-

RSA successfully encrypts and decrypts the message. The original message 5 is recovered

Comparison of classical and Modern Encryption Techniques.

Given plaintext: TRANSFERFUNDS

caesar cipher :-

key = 3

Formula: $c = (P+3) \bmod 26$

Plaintext :- TRANSFERFUNDS

ciphertext :- WLUDQVIHUIXQGV

Playfair cipher :-

key : MONARCHY

key matrix :

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Plaintext pairs: TR AN SF ER FU ND SX

Ciphertext: ZD RA PI KM EV RY XA

Final chiphertext : ZDRAPIKMEVRYXA

Vigenere cipher :-

Plaintext : TRANSFERFUNDS

Key : BANK

Repeated keys : BANKBANKBANK

Formula : $c = (P+K) \bmod 26$

ciphertext : URNXTFRBGUANT

AES-256-GCM :-

* Generate a random 256-bit secret key.
* Generate a unique nonce
* Encrypt the financial information using AES-256-GC
* Generate an authentication tag.
* Send the encrypted data and authentication tag to the receiver
* verify the tag before decrypting the data.

RSA-OAEP :-

* Generate RSA public and private keys
* Use the receiver's public key to encrypt the AES key using RSA-OAEP
* send the encrypted AES key to the receiver.

* The receiver uses the private key to recover the AES key.
* Use AES-256-GCM to decrypt the financial data.

Comparison :-

Algorithm	Main purpose	Security
Caesar	Basic letter shifting	very weak
Playfair	Pair-based encryption	weak
Vigenere	multiple letter shifts	weak
AES-256-GCM	secure data encryption	strong
RSA-OAEP	secure key encryption	strong with suitable key sizes and Padding.

Conclusion :- AES-256-GCM combined with RSA-OAEP is suitable for protecting confidential financial information. classical ciphere are mainly useful for learning encryption concepts.